



ITSIMPLERA SOLUTIONS

INTERNSHIP AI/ML

WEEK 04

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Week 4 Internship Report

Unsupervised Learning & Customer Segmentation

Task: Customer Segmentation using K-Means and Hierarchical Clustering
Dataset: [Credit Card Dataset for Clustering \(Kaggle\)](#) 8,950 customers, 17 behavioral features

1. Objective

This week's task moves from supervised to unsupervised learning. Instead of predicting a known target, the goal is to discover natural groupings within customer credit card behavior, balance, purchases, cash advance usage, and payment history, using clustering algorithms. These groupings can support real business decisions like targeted marketing, credit risk management, and personalized services.

2. Dataset Overview

- **Rows:** 8,950 credit card holders
- **Features:** 17 numeric behavioral features after dropping CUST_ID (identifier, not a feature)
- **Time span:** 6 months of behavioral data
- **Missing values found:**
 - CREDIT_LIMIT: 1 missing value
 - MINIMUM_PAYMENTS: 313 missing values

```
Dtypes:
  CUST_ID                str
  BALANCE                float64
  BALANCE_FREQUENCY      float64
  PURCHASES              float64
  ONEOFF_PURCHASES       float64
  INSTALLMENTS_PURCHASES float64
  CASH_ADVANCE           float64
  PURCHASES_FREQUENCY    float64
  ONEOFF_PURCHASES_FREQUENCY float64
  PURCHASES_INSTALLMENTS_FREQUENCY float64
  CASH_ADVANCE_FREQUENCY float64
  CASH_ADVANCE_TRX       int64
  PURCHASES_TRX          int64
  CREDIT_LIMIT           float64
  PAYMENTS               float64
  MINIMUM_PAYMENTS       float64
  PRC_FULL_PAYMENT       float64
  TENURE                 int64
dtype: object
```

	CUST_ID	BALANCE	BALANCE_FREQUENCY	PURCHASES	ONEOFF_PURCHASES	INSTALLMENTS_PURCHASES	CASH_ADVANCE	PURCHASES_FREQUENCY	ONEC
0	C10001	40.900749	0.818182	95.40	0.00	95.4	0.000000	0.166667	
1	C10002	3202.467416	0.909091	0.00	0.00	0.0	6442.945483	0.000000	
2	C10003	2495.148862	1.000000	773.17	773.17	0.0	0.000000	1.000000	
3	C10004	1666.670542	0.636364	1499.00	1499.00	0.0	205.788017	0.083333	
4	C10005	817.714335	1.000000	16.00	16.00	0.0	0.000000	0.083333	

3. Data Preprocessing

3.1 Handling Missing Values

- CREDIT_LIMIT: the single missing row was dropped, losing 1 out of 8,950 customers has no meaningful effect on the analysis.
- MINIMUM_PAYMENTS: imputed using the median rather than the mean, since this is a skewed financial column and the median is more robust to outliers.

After these steps, 0 missing values remained across the dataset.

```
BALANCE 0
BALANCE_FREQUENCY 0
PURCHASES 0
ONEOFF_PURCHASES 0
INSTALLMENTS_PURCHASES 0
CASH_ADVANCE 0
PURCHASES_FREQUENCY 0
ONEOFF_PURCHASES_FREQUENCY 0
PURCHASES_INSTALLMENTS_FREQUENCY 0
CASH_ADVANCE_FREQUENCY 0
CASH_ADVANCE_TRX 0
PURCHASES_TRX 0
CREDIT_LIMIT 1
PAYMENTS 0
MINIMUM_PAYMENTS 313
PRC_FULL_PAYMENT 0
TENURE 0
dtype: int64
Missing values remaining: 0
```

3.2 Feature Scaling

All 17 features were scaled using StandardScaler (mean 0, standard deviation 1) before clustering.

Scaling is mandatory for K-Means and hierarchical clustering because both rely on Euclidean distance to group similar points. Raw features in this dataset differ wildly in scale, BALANCE and PAYMENTS are in the thousands, while PURCHASES_FREQUENCY and BALANCE_FREQUENCY range between 0 and 1. Without scaling, the large-magnitude columns would dominate the distance calculation, and the frequency/ratio features, often just as informative for telling customers apart — would barely influence the clustering.

```
(8949, 17)
[[-0.73205404 -0.24988139 -0.4249337 -0.35695651 -0.34911414 -0.4668054
 -0.80664908 -0.67871619 -0.70740935 -0.67529446 -0.47608293 -0.51138082
 -0.96037969 -0.52902627 -0.30242676 -0.52558844  0.36054128]
 [ 0.78685815  0.1340494 -0.4695839 -0.35695651 -0.45460688  2.6054375
 -1.22192823 -0.67871619 -0.91708951  0.57394905  0.11003155 -0.59184052
  0.68860056  0.81854607  0.0974534  0.2341587  0.36054128]]
```

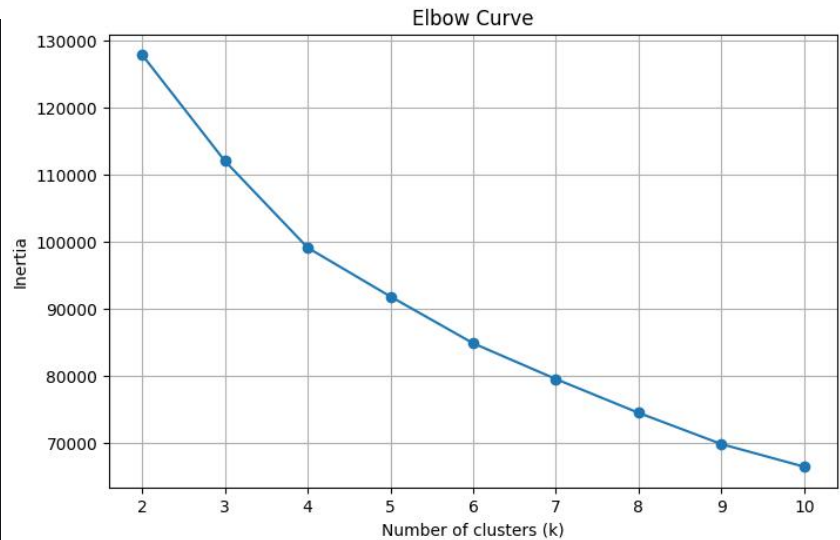
4. Part 1: K-Means Clustering

4.1 Choosing the Optimal k

K-Means was run for $k = 2$ through 10, recording inertia and silhouette score at each step.

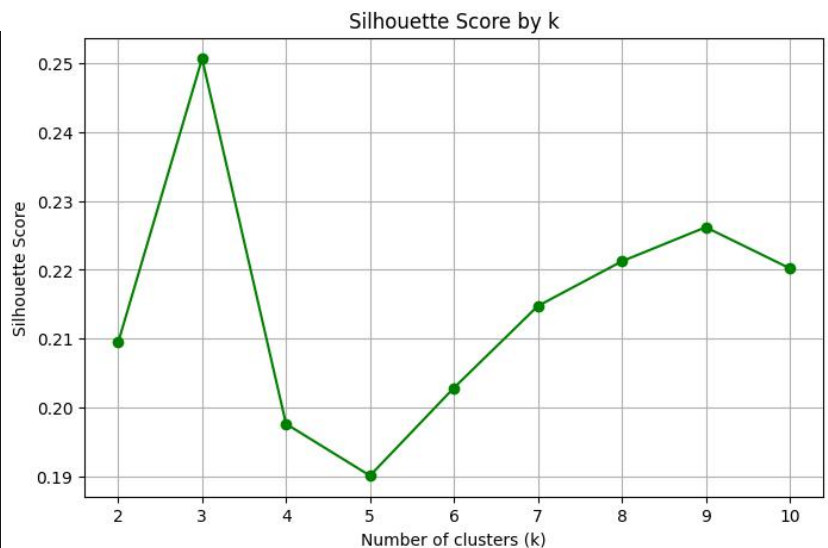
Elbow method: Inertia decreases sharply through $k=4$, then flattens, the drop from $k=3 \rightarrow 4$ (~12,908) is notably larger than the drop from $k=4 \rightarrow 5$ (~7,267), marking the elbow.

```
k=2, inertia=127772.93
k=3, inertia=111962.46
k=4, inertia=99054.06
k=5, inertia=91787.10
k=6, inertia=84819.87
k=7, inertia=79491.72
k=8, inertia=74455.54
k=9, inertia=69805.50
k=10, inertia=66433.78
```



Silhouette method: Peaks sharply at $k=3$ (0.2506), dips at $k=4-5$, then rises gradually toward $k=9$ without beating $k=3$.

```
k=2, silhouette=0.2095
k=3, silhouette=0.2506
k=4, silhouette=0.1977
k=5, silhouette=0.1902
k=6, silhouette=0.2029
k=7, silhouette=0.2148
k=8, silhouette=0.2212
k=9, silhouette=0.2262
k=10, silhouette=0.2202
```



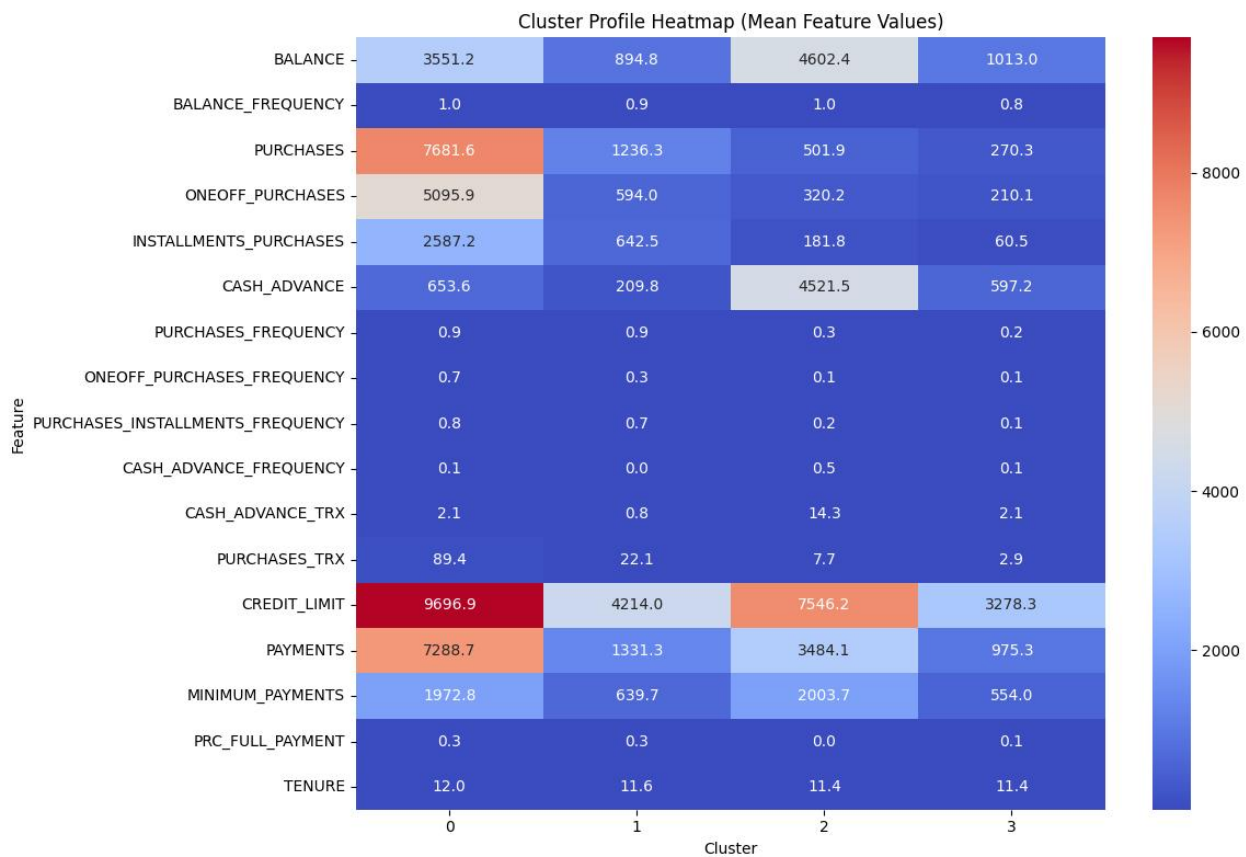
Conclusion: The two methods don't fully agree, silhouette technically favors $k=3$, while the elbow points to $k=4$. Overall silhouette scores are low across the board (all under 0.26), suggesting these customer segments aren't sharply separated, expected for real behavioral data, which tends to sit on a spectrum rather than form tight, distinct groups.

k=4 was selected as the final choice: it's supported by the elbow, its silhouette score (0.1977) is still reasonable relative to the range, and it produces enough distinct segments for meaningful business interpretation without over-fragmenting the customer base.

4.2 Final Clustering Results

```
Cluster
3      3977
1      3366
2      1197
0       409
Name: count, dtype: int64
```

4.3 Cluster Profiling



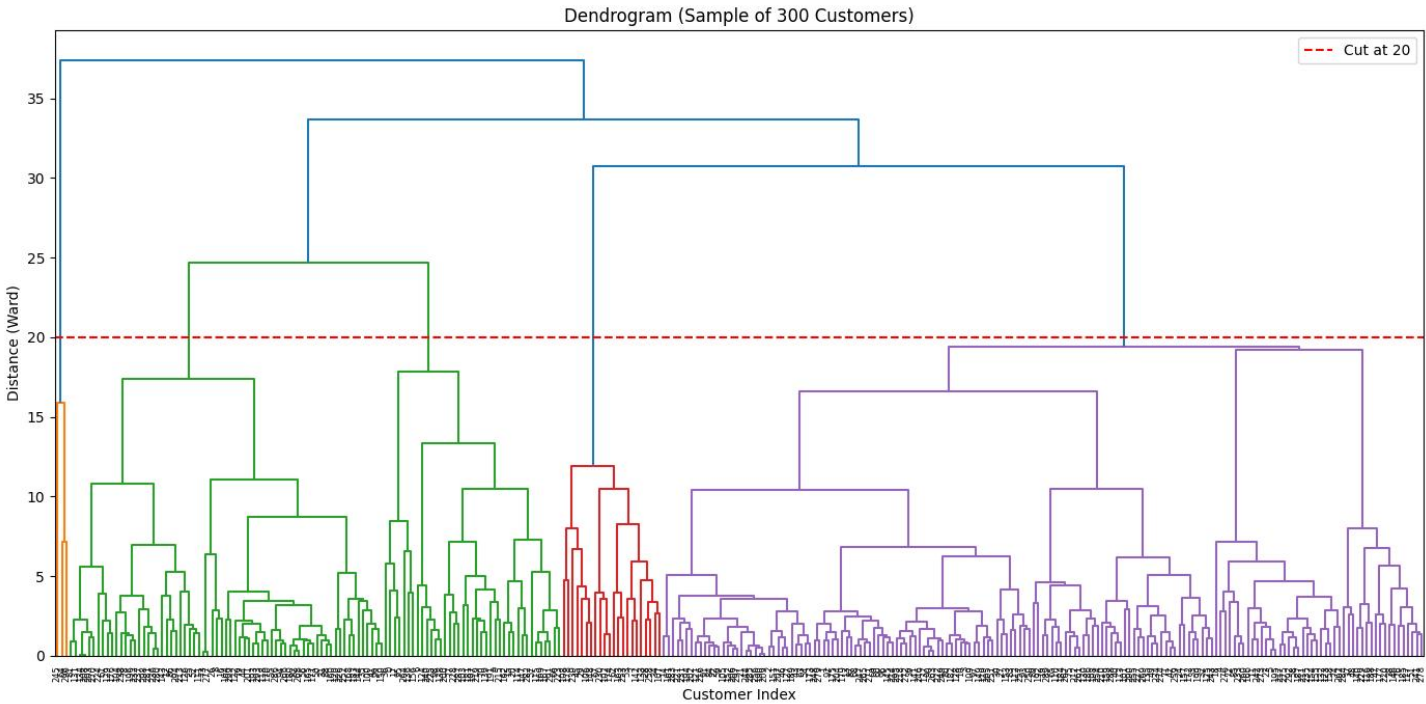
Cluster	Profile
0	Premium spenders: highest purchases (7,681.6), one-off purchases, installment purchases, credit limit (9,696.9), and payments (7,288.7); purchases almost every month; most valuable segment
1	Regular transactors: moderate spend, high purchase frequency (0.89), lowest cash advance frequency (0.04); largest "healthy" segment
2	Cash advance users: low purchases (501.9) and purchase frequency (0.29), but highest cash

	advance amount (4,521.5) and frequency (0.5); highest-risk segment
3	Low-activity customers: lowest activity across nearly every feature; largest group overall; candidates for re-engagement

5. Part 2: Hierarchical Clustering

5.1 Method

Agglomerative Hierarchical Clustering (Ward linkage) was applied to a random sample of 300 rows from the scaled dataset (full-dataset hierarchical clustering is computationally expensive at scale).



The dendrogram was cut at distance = 20, chosen by inspecting the tree and confirming this height splits it into exactly 4 branches, matching k=4 from Part 1.

5.2 Comparison with K-Means

AgglomerativeClustering (n_clusters=4) was applied to the same 300-row sample, and results were compared against the K-Means labels for those same customers using a crosstab.

Hierarchical	0	1	2	3
KMeans				
0	10	3	0	0
1	85	0	30	1
2	3	0	14	21
3	10	0	123	0

Agreement: Both methods clearly separate cash-advance-heavy customers and identify a large low-activity segment, the two most behaviorally distinct groups.

Disagreement: Hierarchical clustering blurs the line between "regular transactors" and "low-activity" customers, mixing them into overlapping branches where K-Means keeps them separate. Hierarchical clustering also split off a very small (3-customer) sub-cluster of premium spenders rather than keeping that segment intact as one group.

5.3 Recommendation

K-Means is the recommended method for real business use. It scales to the full ~9,000 customers (hierarchical clustering only ran on a 300-row sample due to computational cost), produces more consistent and evenly-sized segments, and is simpler to explain to non-technical stakeholders such as marketing or risk teams. Hierarchical clustering remains useful for exploratory analysis, the dendrogram visually shows how sub-groups nest and merge, but turning it into clean business segments requires more manual judgment.

Conclusion

Segmenting customers by behavior rather than demographics reveals four clear, actionable groups: a small premium segment worth retention efforts, a large stable base of regular users, a higher-risk cash-advance-dependent group, and a large inactive segment that represents a re-engagement opportunity. K-Means, at $k=4$, gives the clearest and most business-usable version of this segmentation, while hierarchical clustering served as a useful cross-check on structure.